

**On the Time-Series of Expected Portfolio Returns;
Fama & French's (1993) Three-Factor Model**

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ABSTRACT

The exact specification of the three-factor model of Fama & French (1993) has eluded and defied even its own creators. Fama & French (1996) try to juxtapose the specification of their *ad hoc* model in the context of the ICAPM and APT framework. However, the evidence that has been produced is not sufficiently conclusive. In this paper, we derive an exact specification of the three-factor model of Fama & French (1993) irrespectively of the assumptions that underline the ICAPM and APT frameworks.

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Introduction

Fama & French (1993) building upon inferences drawn from Fama & French (1992), developed an *ad hoc* three-factor model for capturing the expected portfolio returns. Much of the subsequent rationalisation of the model has used a statistic developed by Gibbons, Ross & Shanken (1989); for an application of this statistic see Fama & French (1996). Nevertheless, neither the analysis of Gibbons, Ross & Shanken (1989) nor efforts by Fama (1996, 1998) have provided a convincing explanation of the exact specification of the model or of its coefficients. Moreover, there is also the fundamental question of whether the three-factor model of Fama & French (1993) is embodied within Merton's (1973) ICAPM or Ross' (1977) APT. In this paper, we derive the exact specification of the model irrespective of the ideas that underline the ICAPM and APT frameworks, that is, hedging and arbitrage, respectively.

I) Preliminary analysis

Sharpe (1964) derived the capital asset pricing model (CAPM) that determines the expected return on an individual stock, thus:

$$E(R_i) = R_F + \beta_i (E(R_M) - R_F) \quad (1.1)$$

Where:

$E(R_i)$: the expected return on a stock i

R_F : the return on the risk-free asset

β_i : the market beta, the systematic risk of a stock i

$E(R_M)$: the expected return on the market equilibrium portfolio

In terms of the expected return on a portfolio P , the CAPM is written as follows:

$$E(R_P) = R_F + \beta_P (E(R_M) - R_F) \quad (1.2)$$

Where:

$E(R_p)$: the expected return on a portfolio P

β_p : the market beta of portfolio P

Jensen (1968) by incorporating the single-index model into the CAPM, generalised the CAPM as follows:

$$R_p = R_F + \beta_p (R_M - R_F) + u \quad (1.3)$$

And thus:

$$R_p - R_F = \beta_p (R_M - R_F) + u \quad (1.4)$$

Where:

R_p : the realised return on a portfolio P

R_M : the realised return on the market equilibrium portfolio

u : an error term with a mean of zero and constant variance

Jensen's (1968) model is formulated in realised returns and includes an error term, thus it can be used for testing the empirical validity of the CAPM. More precisely, if the CAPM holds, in the following time-series ordinary least squares (OLS) regression model, the fitted intercept ($\hat{a}_0$) should not be statistically different from zero:

$$R_p - R_F = a_0 + a_1 (R_M - R_F) + e \quad (1.5)$$

Where:

a_0, a_1 : the coefficients to be fitted

e : an error term with mean of zero and constant variance

If the fitted intercept of equation (1.5) is statistically different from zero, it represents an average component of the portfolio P which is not explained by the return on portfolio M . (When equation (1.5) is fitted, portfolio M is a proxy market equilibrium portfolio, such as FTSE100 and S&P500, because the true market equilibrium portfolio identified by the CAPM framework is not observable.) Such empirical results cast serious doubt on the empirical validity of the CAPM, or following Roll (1977), on the mean-variance efficiency of the proxy market equilibrium portfolio used at the RHS of equation (1.5).

However, a significant constant term could be the result of an omitted variable bias. The statistical significance of the intercept could be reduced by adding variables at the RHS of equation (1.5) until the intercept term of the “expanded” model was not statistically different from zero. Thus, the expected return on a portfolio P could be written as a linear function of the expected return on other portfolios. However, to claim that such additional portfolios act as proxy variables for risk cannot be supported simply by their statistical significance.

Consider two portfolios P_1 and P_2 , with realised returns equal to R_1 and R_2 , respectively. Fama & French (1993) construct two such portfolios from four other portfolios, more formally portfolios A , B , C and D . The realised returns on portfolios P_1 and P_2 are defined as the difference between the realised returns on two other portfolios, that is, $R_1 = R_A - R_B$ and $R_2 = R_C - R_D$. This is equivalent to purchasing portfolios A and C , and shorting portfolios B and D .

The augmentation of the RHS of equation (1.5) with the realised returns on two other portfolios leads to the following time-series OLS regression model:

$$R_p - R_F = c_0 + c_1 (R_M - R_F) + c_2 R_1 + c_3 R_2 + \varepsilon \quad (1.6)$$

Where:

ε : an error term with a mean of zero and constant variance

Fama & French (1993) estimate the coefficients of this model using time-series OLS regression analysis. Hence, when the average values of each of the independent variables, that is, their expected values, are inserted into the estimated/fitted time-series OLS regression model, that is equation (1.6), the following relationship is obtained:

$$E(R_p - R_F) = c_0 + c_1 E(R_M - R_F) + c_2 E(R_1) + c_3 E(R_2) \quad (1.7)$$

Further, if:

$$c_0 = 0$$

Then the structure of equation (1.7) becomes the three-factor model of Fama & French (1993, 1996):

$$E(R_p - R_F) = c_1 E(R_M - R_F) + c_2 E(R_1) + c_3 E(R_2) \quad (1.8)$$

Next, we demonstrate how we can obtain equation (1.8) from properties relating to OLS regression model (1.5) and from the mean-variance frontier constructed from two portfolios such as portfolios P_1 and P_2 mentioned earlier.

II) Main analysis

Fitting the OLS regression model (1.5) we obtain:

$$E(R_p - R_F) = \hat{a}_0 + \hat{a}_1 E(R_M - R_F) \quad (2.1)$$

$$\sigma_p^2 = \hat{a}_1^2 \sigma_M^2 + \sigma_e^2 \quad (2.2)$$

The terms $\hat{a}_0$ and σ_e^2 from equations (2.1) and (2.2), respectively, are the fitted intercept and the variance of the error term of OLS regression model (1.5). Their values can be plotted as a “portfolio” in a variance-mean space as point $(\sigma_e^2, \hat{a}_0)$. The

essence of the model developed in this section is that the aforementioned “portfolio”, that is, the unexplained part of the OLS regression model (1.5) can be “explained” as a linear combination of two other portfolios. These two portfolios do not have to be particular embodiments of risk other than each having risk premia.

We now return to the idea that two portfolios P_1 and P_2 can be combined into a portfolio in order to create a frontier (opportunity set) of risks and expected returns. If these combinations are plotted in a variance-mean space, then a frontier in the shape of a parabola is obtained; see Merton (1972) for a formal proof. Thus, the variance and the mean return of any point on the frontier will be attainable by combining portfolios P_1 and P_2 in a particular combination. The frontier is plotted in figure I.

Now consider point $(\sigma_e^2, \hat{a}_0)$, the “portfolio” derived from equations (2.1) and (2.2) which can be plotted in the variance-mean space in figure I. There are three possible cases of where point $(\sigma_e^2, \hat{a}_0)$ can be located. One case is outside of the frontier (case A), one on it (case B), and the final one inside the frontier (case C). In figure I, the risk-free asset which has return and risk (variance) equal to R_F and zero, respectively, is also shown. A capital allocation line can be constructed from point $(0, R_F)$, through case A to intersect the frontier at point $(\sigma_p^{*2}, E(R_p^*))$, where σ_p^{*2} and $E(R_p^*)$ are the variance and the expected return, respectively, of the frontier portfolio at the intersection point. For ease of exposition, the three different possible cases, more formally A, B and C, are all shown on the same capital allocation line.

In the model, the slope and position of the capital allocation line are determined by the location of point $(\sigma_e^2, \hat{a}_0)$ and the return on the risk-free asset R_F . Thus, equations (2.1) and (2.2) along with the risk-free rate determine where the capital allocation line crosses the frontier. It can be seen that case B has variance $\sigma_e^2 = \sigma_p^{*2}$ and mean $\hat{a}_0 = E(R_p^*)$, that is, it is equivalent to a portfolio constructed from the two portfolios P_1 and P_2 . For the other two cases, this does not hold and so there would be an unexplained component that would manifest itself in a positive or negative intercept term (c_0) and in the error term (ε) in equation (1.6).

*** INSERT FIGURE I ABOUT HERE***

Given any of the three cases of point $(\sigma_e^2, \hat{a}_0)$, the expected return on any portfolio that lies along the CAL can be seen to be:

$$E(R_{CAL}) = wE(R_p^*) + (1-w)R_F \quad (2.3)$$

Where:

$E(R_{CAL})$: the expected return on any portfolio located along the CAL

$E(R_p^*)$: the expected return on the frontier portfolio where the CAL intersects the frontier

w : a scalar

Any of the three cases of point $(\sigma_e^2, \hat{a}_0)$ can be considered to be a portfolio constructed on a CAL and so $\hat{a}_0$ can replace $E(R_{CAL})$ in equation (2.3), thus:

$$\hat{a}_0 = wE(R_p^*) + (1-w)R_F \quad (2.4)$$

Next, we write the expected return on the portfolio at the intersection point between the CAL and the frontier as a linear combination of the expected return on portfolios P_1 and P_2 , that is:

$$E(R_p^*) = \chi E(R_1) + (1-\chi)E(R_2) \quad (2.5)$$

Where:

χ : a scalar

Substituting equation (2.5) into equation (2.4), it follows that $\hat{a}_0$ can be written as a linear function of the expected return on portfolios P_1 and P_2 and the return on the risk-free asset, thus:

$$\hat{a}_0 = \varsigma_1 E(R_1) + \varsigma_2 E(R_2) + \gamma R_F \quad (2.6)$$

Where scalars ς_1, ς_2 and γ in equation (2.6) are combinations of scalars w and χ shown in equations (2.3) and (2.5) respectively, that is:

$$\left. \begin{aligned} \varsigma_1 &= w\chi \\ \varsigma_2 &= w(1-\chi) \\ \gamma &= (1-w) \end{aligned} \right\} \quad (2.7)$$

Substituting equation (2.6) into the RHS of equation (2.1) and rearranging we obtain:

$$E(R_P - R_F) = \hat{a}_1 E(R_M - R_F) + \varsigma_1 E(R_1) + \varsigma_2 E(R_2) + \gamma R_F \quad (2.8)$$

This can be rearranged in the form of a time-series OLS regression model with four slope coefficients but no constant term, that is:

$$R_P - R_F = a_1 (R_M - R_F) + \varsigma_1 R_1 + \varsigma_2 R_2 + \gamma R_F + e \quad (2.9)$$

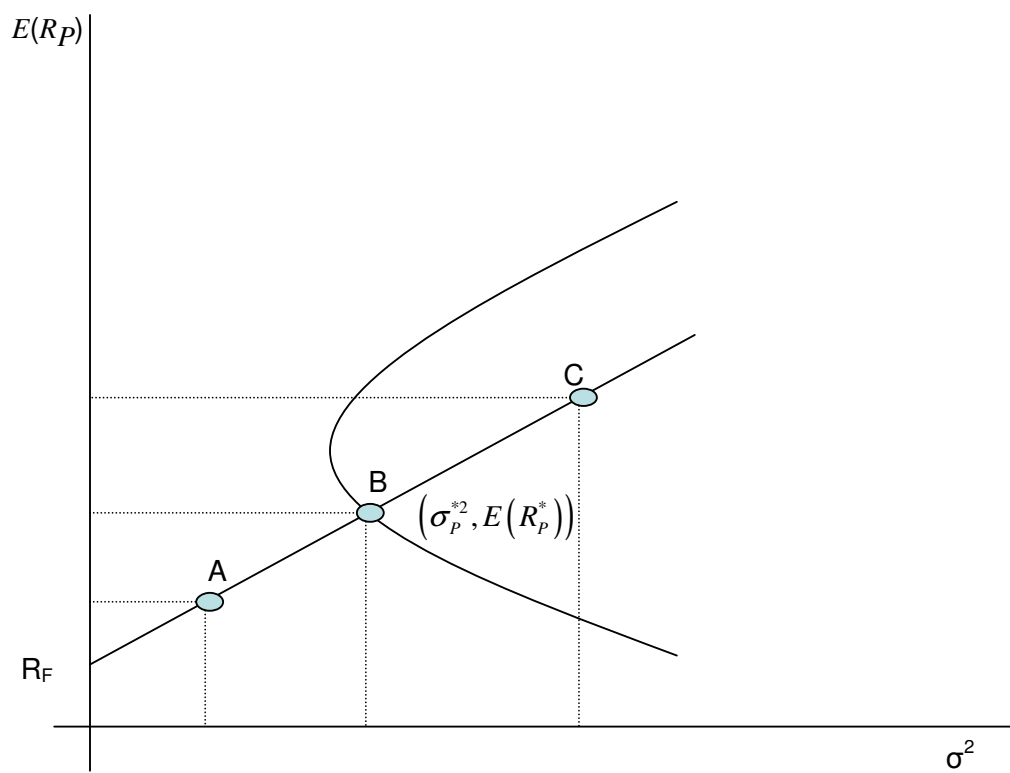
Equation (2.9) therefore shows that the unexplained constant term in equation (1.5) can be expressed in terms of two portfolios and the risk-free asset. However, the idea of an expected risk-premium being a function of the risk-free rate is not only counterintuitive but also contradicts the principle of the separation of the risk premium from the risk-free rate.

III) Conclusions

If $w = 1$, then from equation (2.4), we can see that $\hat{a}_0 = E(R_p^*)$ (and consequently $\sigma_e^2 = \sigma_p^{*2}$) and thus point $(\sigma_e^2, \hat{a}_0)$ is located on the frontier derived from portfolios P_1 and P_2 , that is, case B. Moreover, if $w = 1$, then from equation (2.7) it follows that $\gamma = 0$ and so equation (2.8) becomes the three-factor model of Fama & French (1993) derived earlier as equation (1.8). However, equation (2.7) also implies that if $w = 1$ then $\varsigma_1 + \varsigma_2 = 1$. Thus, the slope coefficients of portfolios P_1 and P_2 in Fama & French's (1993) three-factor model would sum to unity. That $w = 1$ is an implicit restriction in the three-factor model of Fama & French (1993) when estimated using time-series OLS regression analysis. Thus, in any estimation of the model attention should be paid to the statistical significance of the constant term. If it is significant, the implication of the OLS regression model (1.6) is that the constant term indicates omitted variable bias due to the exclusion of the risk-free rate. In such a circumstance the model makes no sense at all. If the constant term is not significant, then $\gamma = 0$ in equation (2.9) and the three-factor model of Fama & French (1993) is the result.

Figure I.

The Two Portfolio Frontier and the Three Cases (A, B, C) of the $(\sigma_e^2, \hat{a}_0)$ Portfolio.



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